

Analyzing Thematic Alignment in World Journal of Orthopedics

Alessandro Macchi

Natural Language Processing
MSc in Data Science for Economics
Università degli Studi di Milano

Abstract

This paper presents a reusable NLP pipeline for quantifying the thematic alignment between scientific article abstracts and a journal’s stated Aims & Scope. Applied to the *World Journal of Orthopedics*, the pipeline collects article abstracts from PubMed and represents both abstracts and the Aims & Scope document as dense vector embeddings using SentenceTransformer (`all-MiniLM-L6-v2`), with a chunk-averaging procedure to handle long texts. Alignment is measured through cosine similarity. Applied to a corpus of 1100 abstracts published between 2015 and 2025, the pipeline yields a mean alignment score of 0.343, a median of 0.342, and a standard deviation of 0.080, suggesting moderate thematic coherence. A z-score outlier rule identifies 47 unusual articles: 20 high-alignment outliers and 27 low-alignment outliers. In addition, BERTopic is used as a complementary diagnostic tool to inspect the latent thematic structure of the corpus over time. Overall, the results suggest that sentence embeddings, cosine similarity and topic modeling can provide an interpretable exploratory framework for studying editorial coherence, thematic drift and outliers in a journal’s publication history.

Keywords: thematic alignment, scientific journals, sentence embeddings, cosine similarity, topic modeling, outlier detection

1 Introduction

The objective of this project is to create a reusable pipeline to assess whether PubMed journals are thematically aligned with their own Aims & Scope. The underlying idea is that the Aims & Scope document can be treated as a reference representation of a journal’s editorial focus, while published article abstracts can be compared against this reference to quantify their degree of alignment. Prior work has demonstrated

that dense vector embeddings derived from pre-trained language models can capture semantic similarity between texts more effectively than keyword overlap [1], motivating the approach adopted here. In this specific case, the analysis focuses on the World Journal of Orthopedics, an open-access, peer-reviewed journal indexed in PubMed. This choice is motivated by the availability of a clearly defined Aims & Scope statement¹ and by the possibility of collecting structured article metadata and abstracts through PubMed. However, the pipeline can be easily reapplied to any journal in the PubMed database. The semantic alignment score is computed using a pre-trained SentenceTransformer model, following the Sentence-BERT approach in which texts are encoded as dense vector representations and compared through cosine similarity [1]. The report presents the methodology used to transform the Aims & Scope and article abstracts into machine-readable representations, compute alignment scores and analyze their distribution over time. The results are discussed both quantitatively, through summary statistics and visualizations, and qualitatively, through the inspection of individual articles, allowing to evaluate the proposed implementation and identify the main limitations of the approach.

2 Methodology

2.1 Research Question

This project investigates to what extent article abstracts in the World Journal of Orthopedics show semantic similarity to the journal’s Aims & Scope. Specifically, the analysis examines whether alignment changes over time, identifies outlier articles with respect to the overall score distribution and uses BERTopic as a complementary tool to inspect the thematic structure of the corpus and support the interpretation of results.

2.2 Data Collection and Preprocessing

The article corpus was collected from PubMed using the World Journal of Orthopedics. The pipeline retrieves article metadata and abstracts and stores them in a structured CSV file. The analysis focuses on title, abstract and publication year, while other metadata such as PMID, DOI, authors, keywords, and journal name are preserved when available. The Aims & Scope text was collected manually from the journal website and used as the benchmark for thematic comparison. The final dataset contains 1,100 abstracts published between 2015 and 2025. Before computing alignment scores, records are validated by retaining only articles with non-missing title, abstract and publication year. Publication years are converted to integers, very short abstracts are removed and duplicates are dropped using title and abstract. Preprocessing is intentionally conservative: whitespace is normalized, while original casing and wording are preserved to maintain information useful for the sentence embedding model.

¹<https://www.wjgnet.com/2218-5836/about.htm>

2.3 Text Representation and Long-Text Handling

The main representation method is based on the pre-trained SentenceTransformer model `sentence-transformers/all-MiniLM-L6-v2`. As stated in the model card, inputs longer than 256 word pieces are truncated by default [2]. To avoid losing information through truncation, which would have affect almost 75% of the abstracts, the implementation introduces a chunking procedure before computing embeddings. Each text is tokenized and split into chunks that fit the model’s maximum sequence length, reserving space for the special tokens added by the tokenizer. Each chunk is then embedded independently. The final text representation is obtained by averaging the chunk embeddings [3].

Formally, let d_i be the text of article i and let its token chunks be:

$$d_i = \{c_{i1}, c_{i2}, \dots, c_{im}\}.$$

Each chunk is embedded by the SentenceTransformer model $\mathbf{v}_{ij} = f(c_{ij})$. The final document embedding is then computed as:

$$\mathbf{v}_i = \frac{1}{m} \sum_{j=1}^m \mathbf{v}_{ij}.$$

The same procedure is applied to the Aims & Scope text. Therefore, the alignment score compares the averaged embedding of each article abstract with the averaged embedding of the journal’s Aims & Scope.

2.4 Alignment Score

After obtaining one embedding vector for each article abstract and one embedding vector for the Aims & Scope document, thematic alignment is measured with cosine similarity. For each article i , the alignment score is defined as:

$$\text{Alignment}(i) = \cos(\mathbf{v}_i, \mathbf{v}_s) = \frac{\mathbf{v}_i \cdot \mathbf{v}_s}{\|\mathbf{v}_i\| \|\mathbf{v}_s\|},$$

where $\mathbf{v}_i$ is the final embedding of article i , obtained either directly or through chunk averaging, and $\mathbf{v}_s$ is the final embedding of the Aims & Scope document. Higher values indicate stronger semantic similarity with the journal’s declared thematic focus.

2.5 Outlier Detection and Aggregation

After computing the alignment scores, the pipeline produces descriptive statistics, yearly aggregations and ranked lists of articles.

To identify unusual articles, the project applies a z-score outlier rule with threshold 2.0. An article is labeled as an outlier when its absolute z-score is greater than this threshold:

$$z_i = \frac{|\text{Alignment}(i) - \mu|}{\sigma}$$

where μ is the mean alignment score and σ is the standard deviation of the alignment scores. Articles with $z_i > 2.0$ are considered outliers.

2.6 BERTopic Analysis

BERTopic is used as a complementary topic-modeling component [4]. This step does not replace the alignment score. Instead, it is used to inspect the thematic structure of the abstracts and to support the interpretation of yearly trends and outlier articles.

BERTopic is fitted using the same pre-computed SentenceTransformer abstract embeddings used for the semantic alignment score. The model first reduces the dimensionality of the embedding space using UMAP, then applies HDBSCAN to identify dense clusters of semantically similar documents. Finally, each topic is represented through class-based TF-IDF, which extracts the most distinctive words for each cluster.

In this implementation, BERTopic may assign some documents to topic -1 , which represents the noise cluster. These documents are not necessarily irrelevant articles, but they are abstracts that the density-based clustering procedure does not assign confidently to a stable topic.

3 Experimental Results

This section presents the results obtained after applying the complete pipeline to the *World Journal of Orthopedics*. The main variable used in the analysis is `alignment_score`, previously defined as the cosine similarity between each article abstract and the journal’s Aims & Scope.

3.1 Overall Distribution of Alignment Scores

The final validated corpus contains 1100 article abstracts published between 2015 and 2025. The average embedding-based alignment score is 0.343, with a median of 0.342 and a standard deviation of 0.080. The minimum observed score is 0.035, while the maximum is 0.747. Figure 1 shows that most articles are concentrated around similar alignment values, especially between approximately 0.25 and 0.45.

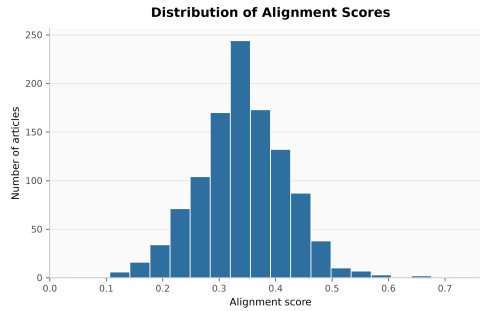


Fig. 1 Distribution of embedding-based alignment scores.

3.2 Alignment Over Time

The yearly aggregation shows moderate variation rather than a clear monotonic trend.

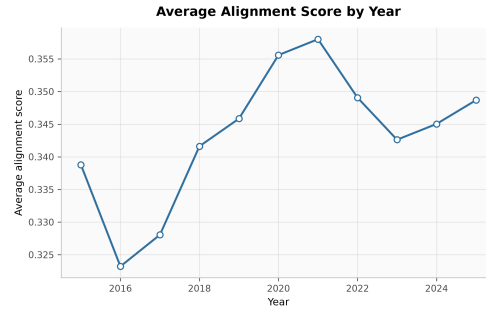


Fig. 2 Average alignment score by publication year.

Figure 2 suggests a mild upward movement in average alignment over time. The alignment score increases between 2016 and 2021, decreases slightly in 2022 and 2023, and then remains close to the central range in 2024 and 2025. The unusual decrease in alignment in the years following the Covid-19 pandemic may reflect the publication of articles related to more unexpected topics, which were not explicitly anticipated in the journal’s Aims & Scope, but also the contribution of the topic class with lower alignment scores as will be discussed in the section about BERTopic. Since no formal statistical trend test is implemented in the project, this result should be interpreted descriptively.

3.3 Outlier Articles

Using the z-score rule with threshold 2.0, the updated pipeline identifies 47 outlier articles. Among them, 20 are high-alignment outliers and 27 are low-alignment outliers. These articles are not automatically errors or off-topic publications, but cases whose semantic similarity to the Aims & Scope is lower than the average article in the corpus.

The highest-scoring article is *New Year’s greeting and overview of World Journal of Orthopedics in 2021*, with an alignment score of 0.747. This result is interpretable because the article explicitly discusses the journal itself. Other highly aligned articles perform a similar function: they summarize broader developments in orthopedics rather than presenting a single narrow research contribution. For example, *Bibliometric analysis of research on the effects of human immunodeficiency virus in orthopedic and trauma surgery*, with a score of 0.675, aims to assess the quantity and quality of peer-reviewed publications in orthopedic journals about HIV. There are also exceptions, such as *Surgical treatment of metastatic bone disease of the distal extremities*, which discusses the treatment of a rare type of bone metastasis and obtains a score of 0.583.

The lowest-scoring article is *Glomus tumors of the fingers: Expression of vascular endothelial growth factor*, with an alignment score of 0.035. This article shares a common feature with other low-scoring articles, such as *New method of local adjuvant*

therapy with bicarbonate Ringer’s solution for tumoral calcinosis: A case report, which obtains a score of 0.085: they use highly specific and technical expressions that may not appear in the Aims & Scope statement. In this situation the metric may fail to fully capture the coherence of these articles with the journal’s scope. For this reason, human inspection remains necessary when interpreting alignment scores.

3.4 BERTopic Analysis

The BERTopic analysis identifies nine non-noise topics, while 267 abstracts are assigned to the noise cluster. Overall, the insights obtained from the topic modeling are coherent with the results derived from the alignment scores.

A first important observation concerns the distribution of abstracts across topics. The first three clusters contain broad articles with average alignment scores; in some cases, these articles are also difficult to assign to a specific thematic group and are therefore labeled as noise. Analyzing this particular cluster, it is possible to notice that there is no real difference between clustered articles and noise articles, both in terms of length of the abstract and in alignment scores; these articles are not assigned because they do not show a distinctive thematic or lexical profile. Taken together, these first three clusters already account for almost 50% of the abstracts. When the noise cluster is also included, this share increases to almost 75% of the entire corpus.

Table 1 Summary of BERTopic clusters and alignment scores.

Topic	Main keywords	Abstracts	Articles (%)	Mean Score
Noise	patients; bone; treatment; results; methods	267	24.27	0.331
G0	fractures; patients; hip; fracture; femoral	239	21.73	0.349
G1	knee; patients; tka; group; bone	193	17.55	0.363
G2	spinal; pain; lumbar; patients; spine	114	10.36	0.322
G3	infection; pji; antibiotic; patients; infections	77	7.00	0.313
G4	covid19; orthopaedic; pandemic; orthopedic; ai	55	5.00	0.403
G5	shoulder; cuff; rotator; tears; repair	53	4.82	0.329
G6	ankle; foot; patients; pain; tendon	44	4.00	0.380
G7	acl; reconstruction; meniscal; graft; ligament	42	3.82	0.309
G8	bone; treatment; lesions; patient; abc	16	1.45	0.310

Because their articles deviate further from the average score, the remaining six clusters offer a more compelling opportunity for analysis. For instance, the highest mean alignment score is observed in 2021, which is also the year in which Group 4 reaches its peak, as shown by Figure 3. This group, characterized by the keywords *covid19*, *orthopaedic*, *pandemic*, *orthopedic*, and *ai*, also has the highest mean alignment score among all groups.

The slow decrease in the mean alignment score observed in 2022 and 2023 can also be interpreted through the topic distribution. In particular, it may be related to a decrease in the number of articles assigned to Group 4, possibly due to a declining interest in COVID-19 and a still limited presence of AI-related articles. At the same time, Group 3 shows a steep increase. This group, defined by the keywords *infection*, *pji*, *antibiotic*, *patients*, and *infections*, contains a large number of articles with low alignment scores. These keywords suggest a highly technical vocabulary, including

acronyms such as PJI, and a focus on surgical complications such as infections and antibiotic treatments. Although these topics are clearly relevant to orthopedic practice, they may be less explicitly represented in the journal’s Aims & Scope because of their multidisciplinary nature within the broader medical field.

Finally, Group 7, characterized by the keywords *acl*, *reconstruction*, *meniscal*, *graft*, and *ligament*, has the lowest mean alignment score. This confirms the initial interpretation of the results: these keywords are clearly coherent with Orthopedics, and a human reader would easily recognize their relevance. However, they receive low alignment scores because they rely on a very specific technical language, even though they remain fully consistent with the journal’s thematic domain.

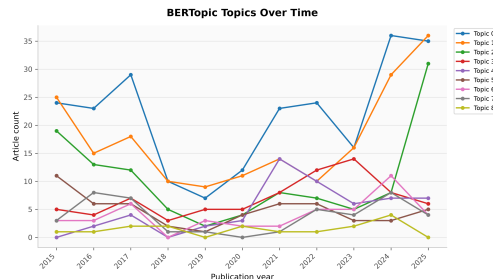


Fig. 3 BERTopic topics over time.

4 Conclusion

This paper presented a reusable NLP pipeline for quantifying thematic alignment between scientific article abstracts and a journal’s stated Aims & Scope. Applied to 1,100 abstracts from the *World Journal of Orthopedics*, the pipeline yields a mean alignment score of 0.343 and identifies 47 outlier articles through a z-score rule. The yearly aggregation reveals a mild upward trend until 2021, followed by a moderate decline in 2022–2023, partially explained by a shift in topic composition captured by BERTopic. Qualitative inspection of outliers confirms a recurring limitation: highly technical or subspecialty-specific vocabulary tends to receive low alignment scores even when the article is clearly relevant to the journal’s domain, as the embedding metric is sensitive to lexical overlap with the Aims & Scope statement. Overall, the pipeline provides an interpretable exploratory framework for studying editorial coherence and thematic drift, but human inspection remains necessary to interpret borderline cases. Future work could address the vocabulary mismatch by experimenting with domain-adapted biomedical language models such as BioBERT or PubMedBERT, or by enriching the reference document with representative abstracts to produce a more robust benchmark for comparison.

AI Usage Disclaimer

For transparency, generative AI tools were used during the development of this project. Anthropic’s Claude Sonnet 4.6 was used in the initial phase to help design the project pipeline and overall structure.

OpenAI’s ChatGPT (GPT-5) was used as a coding and writing assistant, helping to revise parts of the code, improve explanations and draft some sections of the report.

All AI-generated suggestions were reviewed before being included. The author personally ran the code used in the project and checked the outputs reported in the analysis. The research question, methodological choices, interpretation of results and final decisions remain the author’s own responsibility. AI tools were used only as support for planning, implementation, debugging and drafting.

References

- [1] Reimers, N., Gurevych, I.: Sentence-BERT: Sentence embeddings using siamese BERT-networks. In: Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP), pp. 3982–3992 (2019)
- [2] Sentence Transformers: Sentence Transformers: all-MiniLM-L6-v2 Model Card. <https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2>. Accessed: 2026-06-28 (2024)
- [3] OpenAI: Embedding texts that are longer than the model’s maximum context length. https://developers.openai.com/cookbook/examples/embedding_long_inputs. Accessed: 2026-06-28 (2022)
- [4] Grootendorst, M.: Bertopic: Neural topic modeling with a class-based tf-idf procedure. arXiv preprint arXiv:2203.05794 (2022)